DEPARTMENT OF THE NAVY
NAVAL SURFACE WARFARE CENTER
PHILADELPHIA DIVISION
5001 SOUTH BROAD STREET
PHILADELPHIA PA 19112-1403

IN REPLY REFER TO:

J&A Number: 0222-21-321-00.1
Code: 0223
P.R. Number: 21-DAR-006

JUSTIFICATION AND APPROVAL FOR USE OF OTHER THAN FULL AND OPEN COMPETITION

JUSTIFICATION

1. Contracting Activity

   The Naval Sea Systems Command, Naval Surface Warfare Center, Philadelphia Division (NSWCPD), Code 0223.

2. Description of the Action Being Approved

   Award of a Cost-Plus-Fixed Fee (CPFF) Indefinite-Delivery Indefinite-Quantity (IDIQ) contract on a sole source basis to Rolls-Royce Marine North America, Inc. (RRMNA) for Design Agent (DA) Engineering and Technical Services for the Littoral Combat Ship (LCS) Freedom Class, Docking Landing Ship (LSD) 41/49 Class, and Destroyers (DDG) 51 Arleigh Burke Class.

3. Description of Supplies/ Services

   The planned acquisition will provide for highly specialized design agent engineering and technical services for the overall management, development, system testing (inclusive of shipboard testing), troubleshooting, repair, studies, software support, configuration, training support, material support and obsolescence management of the machinery control systems for LCS Freedom Class Machinery Plant Control and Monitoring System (MPCMS) and LCS Freedom Class Control Area Network and Monitoring (CANMAN) Ships Control System, DDG-51 Class Machinery Control System (MCS), and LSD 41/49 Propulsion Logic Management Units (PLMUs).

   The Government’s minimum needs have been verified by the certifying technical and requirements personnel in accordance with FAR 6.303-2(c) (PMS 505, PMS 400D8, PMS 407, PMS 403, and PMS 339).

This acquisition is to satisfy the requirements of the NSWCPD and does not authorize acquisitions for other requirements.

MOPAS Acquisition Strategy Plan No. 0222-21-231-00 covers this acquisition.

The total estimated dollar value of this procurement is $9,818,788 as detailed below.

Table 1 – Acquisition Appropriation Schedule Estimated Dollar Value

Appropriation	FY22	FY23	FY24	FY25	FY26	Total	Percent
OM&N	$1,015,388	$1,017,204	$958,306	$949,333	$969,163	$4,909,394	50%
OPN	$1,015,388	$1,017,204	$958,306	$949,333	$969,163	$4,909,394	50%
Total	$2,030,776	$2,034,408	$1,916,612	$1,898,666	$1,938,326	$9,818,788	100%

The Period of Performance (POP) for this effort will be seventy-two (72) months with a sixty (60) month ordering period.

4. Statutory Authority Permitting Other Than Full and Open Competition

10 U.S.C. 2304(c)(1), Only One Responsible Source and No Other Supplies or Services will satisfy agency requirements, as implemented by the Federal Acquisition Regulation (FAR) Subpart 6.302-1.

5. Rationale Justifying Use of Cited Statutory Authority

This requirement is for highly specialized services that can only be supported by RRMNA as the sole designer and producer of components of the LCS Freedom variant, DDG 51, and LSD 41/49 propulsion and steering related services and equipment.

As the Original Equipment Manufacturer (OEM), RRMNA is the sole designer and fabricator of this equipment, and the only known vendor with the complete technical data and software source code, necessary to provide the associated engineering and technical services. The Navy has contacted RRMNA regarding purchasing the rights to this proprietary technical data. An email request was sent on 8 September 2021 to RRMNA asking to purchase the data rights to these various systems. RRMNA responded on 14 September 2021 via letter, serial number MDK21232, stating that they will not sell the data rights to the Navy. Since the software source code is a required prerequisite in supporting the machinery control systems, and RRMNA is the only known vendor with complete access to the technical data, the government is required to procure such services from RRMNA and does not allow for open competition.

RRMNA has provided PLMU software for twelve LSD Midlife ships and have been involved in development, testing and installation on board all LSD vessels. LSD 41/49 required support for their PLMUs entails software development and testing services. PLMU software was developed specifically for LSD class ships to interface with MCS. This software is proprietary and cannot be modified by any other entity other than the OEM. As a result, Rolls Royce software development services are required.

Functionality of the PLMU is safety critical. A non-functional PLMU could lead to damage to MPDEs (Main Propulsion Diesel Engines). PLMU software is proprietary and requires the OEM to install and modify software as needed.

RRMNA is the sole manufacturer of the CANMAN waterjet system. As such, they are the sole source supplier for parts and services. There are multiple updates planned for this system aboard LCS Freedom class vessels, such as software configuration update to interfacing subsystems, and steering tillers EMI kit. Without this update, there is a risk of losing steerage and situational awareness while underway. Planned updates also require shipboard support for installation and testing of changes, inclusive of grooms and post-system operational verification testing.

RRMNA is the provider of ship specific modeling and analysis for DDG-51 and LCS Freedom class vessels. RRMNA developed ships' models as a sub-contractor to the ship builders (Bath Iron Works/Ingalls and Fincantieri Marinette Marine and Lockheed Martin (LM), respectively) during ship's detailed design and construction phase of the acquisition effort. Prior attempts to secure the developed models directly from RRMNA were unsuccessful.

Due to the retention of data and modeling of the product, RRMNA is the sole industry support agent. In order to find an equivalent service, RRMNA would have to share its information with a prospective lifecycle engineering agent. Barring that occurrence, a need for a replacement system would be evident. A complete development effort would need to take place. This would not only entail replacement of the propulsion logic and regeneration of the Dynamic Report Analysis (DRA); it would require the RRNMA propulsion hardware to be replaced as well, having impact to at least the waterjets, steering control consoles and MPCMS. The new OEM would be required to establish facilities (e.g. office space, labs, storage) for execution of the testing, integration, troubleshooting, configuration management, verification/qualification, and analysis tasks required for the successful lifecycle support of fielded systems. This includes procuring system hardware, test tools, and software components required for development and testing and all shock/vibe/environmental qualification testing. Due to the previous and on-going efforts under LM, RRMNA currently maintains a laboratory in their facility. A new source would need to establish systems engineering baselines for fielded systems based on the current overall technical data packages (TDPs) to support both land-based laboratory and shipboard testing, sea trials, regression testing, integration, configuration management and troubleshooting.

The successful and efficient performance of design agent services in support of the subject systems requires intimate familiarity with the systems. The required in-services engineering support necessitates detailed knowledge of the system configurations and the practices utilized in building and maintaining the systems. RRMNA is the only source who has the requisite knowledge, staffing, and facilities to provide the design agent support to maintain the systems fielded on operational hulls without a learning curve.

Establishing a new DA with the knowledge and infrastructure of RRMNA would result in a substantial duplication of cost to the Government with cost estimates to acquiring these from an alternate source amounting to $13,820,915 in Labor and $4,156,005 in Equipment. Due to the proprietary nature of the PCM software, this may require redevelopment of the mathematical and software models for the platform, still leaving the in-service hulls with no support as NSWCPD is not able to acquire the afore-mentioned software for LCS. DDG-51 has source code and actively maintains the PCM, but the Navy does not have the modeling and simulation software required to support propulsion code changes of a certain magnitude. Changes to the platform, which would affect speed and maneuverability require modifications to these models. In turn, a new DRA and possible source code would be delivered to the Navy. See Table 2 for incurred labor cost breakdown. The new DA would be required to ride ships to gain knowledge required to verify the calculated performance data and rebuild the body of knowledge that RRMNA currently possesses. It is also assumed that RRMNA will sell the pertinent equipment to the new DA. The absence of this material would require intensive research and development in order to produce equipment as similar to the shipboard hardware as possible and mitigation of the remaining risk due to dissimilar equipment. Table 3 shows estimated cost for equipment required to test the subject systems within a laboratory. The estimated cost was derived using the average estimated delay in design, engineering, and logistics support for in-service Freedom variants performing the work for the first time.

Table 2 – Incurred Labor Costs Estimated Dollar Value (NRE)

Labor	Year 1	Year 2	Year 3	Total
Non-Recurring Engineering (NRE)	$3,933,603	$4,684,434	$5,152,878	$13,770,915
Training (10 FTE @ $5,000)	$50,000	$0	$0	$50,000
Total	$3,983,603	$4,684,434	$5,152,878	$13,820,915

Table 3 – Incurred Equipment Costs Estimated Dollar Value (NRE)

Equipment	Price	Quantity	Total
CONTROLLER, SUB SYSTEM (SIM)	$33,525	2	$67,050
MOUNTING KIT	$32,250	2	$64,500
SOFTWARE LICENSE - CANMAN HAND HELD TERMINAL RECORDER SOFTWARE FOR LCS FREEDOM VARIANT	$2,608	6	$15,648
CONTROL CONSOLE (NAVIGATION)	$319,287	1	$319,287
PROPULSION LOGIC MANAGEMENT UNIT (PLMU)	$50,000	2	$100,000
EMPLOYEE SEAT SOFTWARE	$20,000	15	$300,000
NETWORK	$250,000	1	$250,000
LAB CONSTRUCTION LABOR	$3,039,520	1	$3,039,520
Total			$4,156,005

An alternative source to provide continued DA support for systems fielded on the Freedom variant is unlikely. The utilization of in house NSWCPD personnel in place of RRMNA is not an alternative. NSWCPD does not own the data rights to the ship's models, and as such cannot modify/maintain the PCM software. Additionally, NSWCPD does not have the full TDP suite and laboratory environment necessary to successfully support the fielded systems, which would result in an immediate degradation to fielded systems on operational Freedom variants.

Based on the information stated above, RRMNA is the only company capable of readily providing the DA services necessary to sustain the specified systems at the design performance levels in an acceptably affordable manner through the system's end-of-life. Failure to award the IDIQ would jeopardize the lifecycle supportability of current fielded systems and thereby degrade the mission readiness and capability of the subject platforms.

The utilization of in-house NSWCPD personnel in place of industry expertise is not an alternative. NSWCPD personnel do not currently possess the requisite knowledge or skillsets to perform these tasks. NSWCPD would need to stand up as a DA and assume the cost (Table 2) associated with growing the capability. Thus, these services are deemed to be available only from RRMNA for the duration of the sixty (60) month ordering period, as award to any other source would result in substantial costs to the Government associated with establishing a new DA. This cost is not expected to be recovered through competition.

6. Description of Efforts Made to Solicit Offers from as Many Offerors as Practicable

A Request for Information/Sources Sought was posted to Beta.SAM.Gov on 10 February 2021 with a closing date of 19 February 2021. Additionally, a synopsis of this procurement was published to Beta.SAM.Gov on 9 June 2021 with a closing date of 24 June 2021. By the closing date, no contractors had responded to either posting. No additional market research was conducted because it is not practicable, for the reasons discussed in paragraph 5 above, for any other contractor than RRMNA to provide the required services.

7. Determination of Fair and Reasonable Costs

The Procuring Contracting Officer (PCO) has determined that the anticipated cost to the Government for the services covered by the J&A will be fair and reasonable. The contractor will be required to submit a formal cost proposal with certified costs or pricing data and sufficient information to support the accuracy and reliability of the proposal. Experienced technical personnel and the contract specialist, with the aid of necessary field pricing support, will review the contractor's proposal. The PCO will use cost and pricing analysis as the basis for negotiating a fair and reasonable price.

8. Actions to Remove Barriers to Competition

This requirement is not considered to be a follow-on procurement in accordance with FAR 6.302(a)(2)(iii) and DFARS PGI 206.303-2(b)(i)(B). For the reasons set forth in paragraph 5, NSWCPD has no plans at this time to compete future contracts for the services covered by this justification. All requirements that arise under this IDIQ will be reviewed annually by the PCO to ensure that RRMNA is the only company that can satisfy agency requirements.

The technical team and the PCO will assess any potential future sources that may be discovered to determine whether competition for future requirements is feasible.

TECHNICAL AND REQUIREMENTS CERTIFICATION (FAR 6.303-2(c))

I certify that the facts and representations under my cognizance, that are included in this justification and its supporting data, including MOPAS No. 0222-21-321-00, except as noted herein, are complete and accurate to the best of my knowledge and belief.

COMBINED TECHNICAL AND REQUIREMENTS COGNIZANCE:

[redacted] [redacted] [redacted]

Signature: [redacted] Phone No. Date

Technical Division Head (Code [redacted])

LEGAL SUFFICIENCY REVIEW (NMCARS 5206.303(90)):

I have determined that this justification is legally sufficient.

[redacted] [redacted] [redacted]

Signature: [redacted] Phone No. Date

Office of Counsel (Code [redacted])

CONTRACTING OFFICER CERTIFICATION (FAR 6.303-2(b)(12)):

I certify that this justification is accurate and complete to the best of my knowledge and belief.

[redacted] [redacted] [redacted]

Signature: [redacted] Phone No. Date

Contracting Officer (Code [redacted])

APPROVAL BLOCK (FAR 6.304 for Approving Official)

Upon the basis of the above justification, I hereby approve, as Competition Advocate of the Procuring Activity, the solicitation of the proposed procurement(s) described herein using other than full and open competition, pursuant to the authority of 10 U.S.C. 2304(c)(1).

CHIEF OF THE CONTRACTING OFFICE

STRACCIOLINI.ALBERT.1229157244 Digitally signed by STRACCIOLINI.ALBERT.1229157244 Date: 2021.11.02 07:33:51 -04'00' 11/2/21

Signature: [redacted] Date